

Rayuela: Delivering Serious Information Through Playful Interactive Installations

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ABSTRACT

Rayuela is an interactive light installation created with the purpose of using art to draw attention to a serious issue plaguing our society: namely plastic usage and waste, and how its overuse and under recycling end up polluting the oceans and compromising the food chain by disintegrating into microplastics. To create an experience that could be playful enough to attract and engage people, while still subtly conveying a serious message, we chose the game of hopscotch as a backdrop for a playful experience, revealing at each step the journey a plastic bottle goes through during its lifecycle. Accordingly, we named the work Rayuela, which is the Spanish word for hopscotch. We built Rayuela as a modular system consisting of 11 light boxes, each telling a different fact about the plastic bottle's journey. Each time people jumped on a box, it turned the next one on, thus progressively revealing the story. As the work was designed for exhibition in a very large public festival lasting about a month, design considerations ranged from aesthetic and interactive aspects to robustness and reliability. In this paper, we describe the conceptual rationale for Rayuela, its design and building process, and the methods for automated data gathering we implemented. We then present the results of the field study, and discuss preliminary findings, as well as implications for future design.

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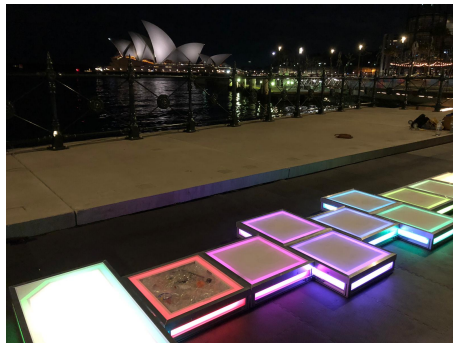


Figure 1: Rayuela: an interactive hopscotch telling the story of a plastic bottle, as exhibited in Vivid Sydney 2019.

CCS CONCEPTS

• **Human-centered computing** → **Interface design prototyping**; *User centered design*; *Activity centered design*.

KEYWORDS

urban media art, storytelling, playfulness, embodied action, design.

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INTRODUCTION

A lot of serious issues plague our society. These issues range from personal issues to global issues, with causes and relationships to day-to-day life often difficult to understand by non-experts. Such complexity **undermines** positive action toward their resolution and recurrence prevention. **It is thus important that people become aware of how their actions may lead to society's issues, so that they can start making changes to their behaviour that can be effective in improving human existence in this world and to preserve the planet we all live on.** Art has been a very powerful tool to raise this awareness, as artists find different ways to express and explain the problems we collectively face, **appealing to raw, tangible feelings, and urging people to enact change.**

Examples of art concerned to this sort of social activism **abound**. Alinta Krauth, for instance, presented a solo exhibition called **Undermine at the Art Laboratory Berlin**, raising awareness to the effect of climate change on animals. She presented an interactive work, in conjunction with 4 other digital works, which through **gamification** allowed people to explore the possibilities of inter-species empathy and **dislodge** the usual subject-object relation to the nonhuman [5]. Likewise, *Clams*, a kinetic sound sculpture by Marco Barotti, generates a **soundscape** and an opening and closing sequence based on real time water quality data. In doing so, it intends to raise awareness about water and plastic pollution [1].

Plastic in the ocean is a growing concern: approximately 21.7 million tonnes enter our oceans every year [4]. In addition to the consequences of pollution to water consumption and wildlife, its effects become even more pernicious - and its resolution desperately urgent - following studies reporting that plastic in the ocean has disintegrated to the point of becoming microplastics, capable of being reintroduced into the food chain [8]. **Moved by this concern, we wanted to raise general awareness about this issue, and prompt people to make a change.**

RAYUELA

Rayuela is an interactive installation originally created by designers Abhiruchi Chhikara, Iskra Iscumlic and Florencia Diniello for Vivid Sydney 2019, an annual light and music festival in Sydney, Australia (Figure 1). It emerged out of a desire to spark discussion about the aforementioned global plastic issue in a way that could be more fun and engaging than traditional educational campaigns. Given the nature of the festival, which has families as a core target audience, we expected a large portion of the participants to be children. We thus decided to adopt embodied action as strategy to grab attention of passers-by, stimulate cognition and foster further action [6]. The chief aim was to tell a story that could use embodied action to empower further action, rather than merely showing statistics that, in other contexts, could have been just easily ignored [7]. Adopting this approach, we decided to tell the story of a plastic bottle, as a simple, easily recognisable and relatable starting point for the subject.

Rayuela's layout is inspired by a well-known children's game: hopscotch (*rayuela* in spanish). The intention was to have the game of hopscotch acting as a hook, creating an intuitive public interaction by relying on the familiar experience of a children's play popularly enacted with chalk on the pavement, in order to lower the entry barrier for participation and prompt people's engagement [3]. Once having their attention, we could then deliver serious information in small chunks, so that they could slowly unravel and easily absorb the information. To that end, and in line with the classical design for a hopscotch, Rayuela was made out of 11 light boxes. Out of those, 10 were storytelling boxes and the last one would provide a conclusion in the form of forward action to address the issue, by displaying alternatives to plastic usage. That way, the structure of the play would support the structure of the storytelling process: each jump (bodily action) would bring new information (reaction), up to the end, when a way forward for the tale was presented. Moreover, we expected that the continuous physical activity could help alleviating the seriousness of the topic and making participants simultaneously relaxed and curious to received each new piece of information.

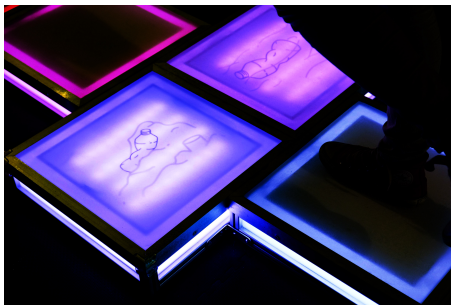


Figure 2: Stencils on the top of the boxes, telling on each step facts about the plastic bottle lifecycle.

DESIGN AND DEVELOPMENT

Each of the 11 boxes is equipped with LED strips and pressure sensors. Once the sensors in one box are activated, they trigger the LEDs in the next box to light up. When the light within a box turns on, it reveals illustrations stencilled on its top (Figure 2). The system is essentially modular, with each box following the same basic construction steps, apart from a few minor differences in electronics and programming, to cater for events associated to their specific position within the players journey. Following this modular design, we decided to build one box first to the point all interactions were working successfully, before scaling its implementation up to multiple modules. We kept the basic design of all boxes consistent for a few reasons:



Figure 3: 11 Identical boxes consisting of a shell with an internal structure to house electronics and activate sensors.

- **Ease and speed of development:** designing one could enable us to know how all of them, and thus the whole system, would look, feel and behave. It also enabled us to take advantage of working in parallel and keep construction time feasible within the short time-frame we had to develop it (around 4 months of part-time work).
- **Ease of maintenance:** if any box failed, troubleshooting and fixing would follow the same steps.
- **Modular:** we could use the boxes interchangeably and in alternative configurations in future deployments.

Each box consists of a metal shell which encloses an internal structure that is used to mount the LED strips and transfer the weight load of participants to activate the pressure sensors underneath (Figure 3). The box is topped with an acrylic panel that has a vinyl sticker with the illustrations stencilled on it, conveying facts about the plastic bottle's journey. The 10th box does not have an illustration instead it houses and displays plastic waste collected in Sydney.

The Shell

Each box consists of a metal frame made out of standard aluminium extrusions. These sort of extrusions are traditionally used for shelving, and therefore are very robust and capable of withstanding high loads. The aluminium extrusions were 30 mm wide square cross sections connected together with standard attachments. A 6mm perspex panel formed the base and 3mm frosted perspex panels were fitted in each side allowing the light from LEDs to shine through. All edges were sealed from the inside with silicon to waterproof the box as it housed electronics. On the top, a 15mm transparent acrylic panel served as a platform for people to stand and jump on. The top acrylic was secured into place with custom made aluminium angles to have a sleek top with no attachments or screws as well as minimise water ingress.

Internal Structure

The internal structure for the box module, made out of MDF, fulfills a variety of functions. These includes securing the LED strips in place, transferring weight load from players onto 2 or 4 pressure sensors depending on the size of the box, as well as creating a subtle outline around the top of the box, clearly distinguishing one step the hopscotch from another. The structure has an upside down T-cross section. The pressure sensors are positioned under the flat base, sandwiched between slices of foam matting. This creates a neat mechanism whereby, when someone steps on the top acrylic, it flexes and pushes the structure against the pressure sensor underneath. The foam matting gives the structure space to move and provides cushioning to protect the sensor, while also ensuring that the structure always touches the top acrylic, thus compensating for minor variations in the finishing of



Figure 4: Pressure sensing was the user input for the installation, which would trigger the next step to light up.



Figure 5: Rayuela was very popular among kids, some participants walked in pairs taking the journey together.

each individual box. All electronics, including an Arduino board and, for some boxes, a SD card, are housed in a plastic container stored in the box's internal cavity.

PRESENCE SENSING AND DATA GATHERING

When the system starts, it **calibrates** the **'idle'** values on the pressure sensors. **The program constantly checks for change in the pressure values from idle.** The pressure from all sensors is checked to cater for off-centre loads. When someone jumps on the step, the pressure difference goes **over a preset threshold value**, calibrated during prototyping testing triggering the light of the next box (Figure 4).

A SD card was added to the first box to record the number of times the box was triggered. Every day the system would start up for 20 minutes, for testing, and then subsequently run for 5 hours: from 6pm to 11pm. We collected data for the first 3 hours of attendance, i.e. 6pm to 9pm, which corresponds to peak attendance time. Although mostly straightforward, there were some relevant challenges concerning data recording and analysis. For example, as Arduino boards do not have an internal clock, only the relative time since startup could be recorded. Since the installation was not on all the time and the internal timer would reset, we needed to make sure that data was recorded with a **timestamp** so we could later on extrapolate days and dates and interpret data correctly. This became harder when we had to perform maintenance work, due to multiple resets making the data analysis cumbersome.

PRELIMINARY RESULTS

A variety of people interacted with Rayuela over the 4 weeks of the festival. **Visitors included people of all ages from all over the world.** It was clearly **very popular among kids** (Figure 5), **who not being worried about how they would look in public, would line up waiting for the festival to kick off and the lights to come on.** One of the most common comments heard from people who spotted the installation was *'is that a hopscotch?'* or *'I think that is a hopscotch'* or *'That's a hopscotch!'* Although most people most likely knew what the work was, it still seemed to have **triggered a sense of excitement and nostalgia on passers-by**, as they **discovered something familiar in an different setting.**

Rayuela was first used by the public during the festival's media night, which happens a few days prior to its launch. On that night, there were no volunteers to direct the participants, and the work was fully open for **spontaneous use.** This made us realise that, **due to the storytelling angle we adopted, we had our installation designed to be unidirectional, in contrast to 'traditional' hopscotch in which people turn around and hop back to the start.** Without any instruction people followed what they knew. They also did **not care about the images being upside down: as we observed, they would simply enjoy hopping around and seeing the boxes light up.**

Children also often wanted to cross from one end to the other as fast as they could, **without skipping a step. They would then be cheered on and celebrated when arriving at the other end.** Admittedly,

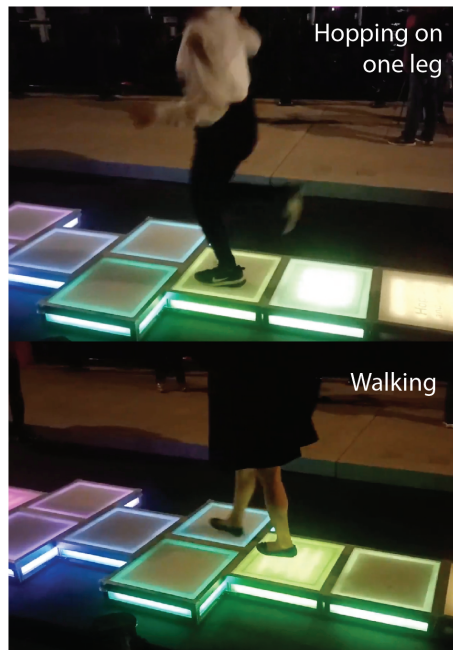


Figure 6: Some people hopped across the installation others chose to walk.

those participants did not pay much attention to the images or, sometimes, even to the changing lights. But then there were those, both children and adults, who did carefully stepped on each box at a time. Some parents would make the journey with their kids walking from one step to another, holding hands. Some people jumped through it with one leg, as it is done traditionally, and others just walked through (Figure 6). People would also recurrently ask friends/partners to take photos and videos of them jumping through, to capture the experience as well as potentially to share on social media.

Based on the data we gather over the 4 weeks, Rayuela logged over 40,000 steps. Even if a small number of people out of the 40,000 learned from this experience and, as a consequence, try to reduce plastic use in their daily lives, it would create a significant impact on the world. Even more so, as their actions and behavioural change can also have a ripple effect in their community and social networks.

CONCLUSION AND FUTURE WORK

With art, what people feel in the experience is most important [2]. Artists and designers should weave evaluation into the design artworks even when the artworks are not specifically for research. It allows us to find out what the artwork achieved, and to establish the social value or worth of the installation [2]. Digital art allows us to embed these evaluations into the artwork more easily than traditional arts. For Rayuela the evaluation systems can be improved further. We collected data that tells us how many people interacted with the installation, yet we could refine it even more by collecting data on the degree of engagement e.g. how much time did people spend on each step. We would also benefit from having a structured data collection system regarding how the audience interacts with the installation. This could be in the form of regular observations sessions during the time of the exhibition, video recordings for processing later in addition to surveys and interviews to gauge what people felt and what they got out of the experience.

From a storytelling perspective, the illustrations for the boxes can be explored further. Some people did not pay attention to the images, only the blinking lights, and jumped through the hopscotch very fast as that matches the traditional mental model for the game. Stopping at steps is not normal in that context. More elaborate and intricate illustrations could perhaps encourage participants to stop and stare at them, hence spending more time on each step. Another option would be to design a two-staged interaction, where even if people crossed the path fast they could feel tempted to come back and have another look to appreciate the detailed visuals on the boxes.

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REFERENCES

- [1] Marco Barotti. 2019. Clams. Retrieved Oct 10, 2019 from <https://www.marcobarotti.com/CLAMS>
- [2] Linda Candy and Sam Ferguson. 2014. *Interactive Experience in the Digital Age: Evaluating New Art Practice*. Springer Publishing Company, Incorporated.
- [3] L. Hespanhol and M. Tomitsch. 2015. Strategies for Intuitive Interaction in Public Urban Spaces. *Interacting with Computers* 27, 3 (May 2015), 311–326. <https://doi.org/10.1093/iwc/iwu051>
- [4] Jenna R. Jambeck, Roland Geyer, Chris Wilcox, Theodore R. Siegler, Miriam Perryman, Anthony Andrady, Ramani Narayan, and Kara Lavender Law. 2015. Plastic waste inputs from land into the ocean. *Science* 347, 6223 (2015), 768–771. <https://doi.org/10.1126/science.1260352> arXiv:<https://science.sciencemag.org/content/347/6223/768.full.pdf>
- [5] Alinta Krauth. 2017. Under-Mine. Retrieved Oct 10, 2019 from <http://alintakrauth.com/undermine/>
- [6] Mariana Lozada and Natalia Carro. 2016. Embodied Action Improves Cognition in Children: Evidence from a Study Based on Piagetian Conservation Tasks. *Frontiers in Psychology* 7 (2016), 393. <https://doi.org/10.3389/fpsyg.2016.00393>
- [7] Jörg Müller, Dennis Wilmsmann, Juliane Exeler, Markus Buzeck, Albrecht Schmidt, Tim Jay, and Antonio Krüger. 2009. Display Blindness: The Effect of Expectations on Attention towards Digital Signage. In *Pervasive Computing*, Hideyuki Tokuda, Michael Beigl, Adrian Friday, A. J. Bernheim Brush, and Yoshito Tobe (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 1–8.
- [8] F.A.O.U. Nations. 2018. *Microplastics in fisheries and aquaculture:: Status of knowledge on their occurrence and implications for aquatic organisms and food safety*. Food and Agriculture Organization of the United Nations. <https://books.google.com.au/books?id=eGp3DwAAQBAJ>